

Uthpala Herath, Ph.D.

COMPUTATIONAL PHYSICIST

Durham, NC

☎ (304) 216-2535 | ✉ uthpala.herath@duke.edu | 🏠 www.uthpala.herath.com | 🌐 uthpala.herath | 📄 Google Scholar

Summary

Innovative computational physicist with **10 years** of experience leveraging research computing and high-performance computing for advanced materials research. Specializes in the development and application of frameworks utilizing beyond-DFT methods to design, characterize, and advance novel materials critical to semiconductors, spintronics, neuromorphic computing, electronics, optics, photovoltaics, and energy applications. Proven leader and mentor in pioneering collaborative projects, securing over **\$1.1 million** in grants, publishing research recognized with **500+ citations**, and developing open-source tools adopted by **400+ global users**. Dedicated to applying advanced computational, problem-solving, leadership, and mentoring skills to spearhead transformative projects in materials research and beyond.

Education

Ph.D. in Computational Condensed Matter Physics

Department of Physics and Astronomy, West Virginia University

Advisor: Dr. Aldo H. Romero

Dissertation: "Development of computational methods for electronic structural characterization of strongly correlated materials: from different ab-initio perspectives"

Morgantown, WV

Aug, 2015 - May, 2022

M.Sc. in Computational Condensed Matter Physics

Department of Physics and Astronomy, West Virginia University

Morgantown, WV

Aug, 2015 - Aug, 2017

B.Sc. in Physics

Department of Physics, University of Peradeniya

Peradeniya, Sri Lanka

Jul, 2010 - Dec, 2014

Research experience

Department of Mechanical Engineering and Materials Science, Duke University

Durham, NC

Postdoctoral Associate | Blum Group (Computational Materials Science)

May, 2022 - Apr, 2025

- Built and maintained scientific CPU/GPU applications and modules on DCC and Timewarp HPC clusters, authored user documentation, and delivered hands-on user training for research and education.
- Developed a novel Δ -SCF method in **FHI-aims** and **ELSI** to discover high-temperature superfluorescence in collaboration with experimental teams, advancing research in quantum computing and optoelectronics, leading to a publication in *Nature*.
- Enhanced materials science databases (**Hybrid³**, **MatD³**, and Springer Materials) by refining critical features for improved functionality.
- Led the integration of relativistic spin-orbit coupling into periodic GWA within the DFT code, **FHI-aims** and conducted comprehensive GWA benchmarks exploring compound semiconductors and heavy-element materials for electronics and energy applications.
- Coordinated a global initiative to enhance **FHI-aims** and **ELSI** usability through code development efforts, including the implementation of a CI pipeline utilizing Docker and integrating cutting-edge eigen solvers and density matrix solvers.

Department of Physics and Astronomy, West Virginia University

Morgantown, WV

Graduate Research Assistant | Romero Group (Computational Condensed Matter Physics)

Aug, 2017 - May, 2022

- Developed the open-source **DMFTwDFT** framework, significantly advancing the study of strongly correlated materials through the many-body Green's functions method, DMFT. Maintained a code repository and a user forum with 40+ global users, with a publication cited 35+ times.
- Led research on the effects of oxygen vacancies and hydrogen doping on metal-insulator transitions and diffusion in rare-earth nickelate perovskites, with implications for neuromorphic computing and microelectronics.
- Developed neural network-based interatomic potentials capturing DMFT-level accuracy for strongly correlated materials, enabling large-scale MD simulations across diverse time and length scales through the interface of **DMFTwDFT** and BLAST frameworks.
- Implemented a symmetry based method to investigate alloying/defects in strongly correlated materials with higher computational efficiency.
- Developed the open-source electronic structure pre/post-processing tool **PyProcar**, maintaining a code repository and a global user forum of 300+ users, with a publication cited 325+ times.
- Developed **MechElastic**, an open-source Python library for calculating elastic properties of materials, maintaining a repository and a user forum of 50+ global users, with a publication cited 120+ times.
- Developed **MDWC**; an open-source Python program for constrained molecular dynamics simulations and **PyChemia**; an open-source Python library for materials structural search through debugging, simulation, and testing.

Department of Physics and Astronomy, West Virginia University

Morgantown, WV

Graduate Research Assistant | Tu Group (Computational Space Physics)

Jan, 2016 - May, 2017

- Developed and applied a HPC C/C++ particle tracer code to perform test particle simulations to quantify the effect of magnetic field line curvature (FLC) scattering on the rapid loss of ring current ions in Earth's magnetosphere.

Arthur C. Clarke Institute for Modern Technologies

Moratuwa, Sri Lanka

Intern (Space Applications Division) | Advisors: Mr. S. Gunasekara, Dr. T. Ranawake

Jan, 2014 - Dec, 2014

- Utilized telescopic data to construct a novel Color-Magnitude Diagram (CMD) of the globular cluster M53, mapping Blue Straggler stars using CCD Aperture Photometry and PSF (Point Spread Function) Fitting Photometry methods with the IRAF (Image Reduction and Analysis Facility) system.

Department of Physics, University of Peradeniya

Peradeniya, Sri Lanka

Undergraduate Research Student | Advisor: Dr. P. Samarasekara

Jan, 2013 - Dec, 2013

- Investigated the traits, synthesis methods, functionalization, and applications of nano-magnets.

Skills

First principles (*ab-initio*) electronic structure calculations

- Expertise in developing and applying electronic structure codes **FHI-aims**, VASP, Siesta, Abinit, Elk, and Quantum Espresso for DFT, DFT+U, molecular dynamics, and many-body Green's functions methods (GWA, DMFT) for materials research
- Skilled in the analysis of electronic, structural, vibrational, magnetic, mechanical, thermal, and optical properties of multi-dimensional materials
- Proficient in studying properties of strongly correlated materials using DMFT

Development and application of materials research methods and tools

- **DMFTwDFT**, **ELSI**, Wannier90, Impurity solvers and QMC methods, Tightbinding Hamiltonians, Phonopy
- Materials discovery, workflows and analysis (**PyProcar**, **PyChemia**, **MechElastic**, Materials Project, Pymatgen, AFLOW, AiiDA, Atomate2)
- Transition state and diffusion analysis (VTST Tools, DiSPy, Nudged Elastic Band (NEB) method)
- Molecular dynamics simulations (**MDWC**, VMD, Jmol, qAgate)
- Graphical visualization (VESTA, gnuplot, Xmgrace, xcrysden, matplotlib)
- Vacancy, defect and alloying methods (Virtual Crystal Approximation (VCA), Coherent Potential Approximation (CPA), Site Occupation Disorder (SOD))

Programming environments

- Python, Fortran, C/C++, Bash, Matlab, Octave, Java, Django, GPU programming (CUDA), Jupyter Notebooks

Data science and machine learning

- Artificial Intelligence (AI), Machine Learning (Supervised, Unsupervised, Reinforcement Learning), Logistic Regression, Multilayer Perceptrons, Deep Learning, Convolutional Neural Networks (CNN), Natural Language Processing (NLP), Image and Text Processing, Predictive Modeling, Computer Vision, PyTorch, scikit-learn, nltk
- Data Mining, Data Cleaning and Manipulation, Data Visualization, Statistical Inference, Graph and Network Analysis, FAIR Data Management
- Database Management Systems (SQL, MongoDB)
- Python libraries for data science (NumPy, SciPy, pandas, NetworkX, regex, matplotlib)

Computational methodologies

- Linux/Unix systems, Research Computing, High-Performance Computing (HPC) experience with major supercomputing clusters (TACC, PSC, NERSC), High-Throughput Computing (HTC), Parallel computing (OpenMP and MPI), Differential solvers and integrators, Numerical analysis algorithms, Object-oriented and procedural programming paradigms, Scientific libraries (LAPACK, BLAS, SCALAPACK, FFTW, GSL)

Version control, CI, and build systems

- Git, Gitlab CI, Travis CI, Docker, Apptainer, CMake, CTest, and Unit testing

Graphics and publishing

- Latex, Markdown, Zotero, Adobe Creative Suite (Photoshop, Premiere, After Effects, Illustrator), Microsoft Office

Languages

- English (Native or bilingual proficiency), Sinhala (Native or bilingual proficiency), Tamil (Elementary proficiency), Pāli (Elementary proficiency)

Teaching & mentorship

Postdoctoral Associate (Duke University)

May, 2022 - Apr, 2025

- Guided students in using HPC clusters for electronic structure calculations with the **FHI-aims** code while providing hands-on training with Python, Fortran, and Bash for advanced materials research.
- Led students in the design and implementation of the **HybriD³** materials database and **MatD³** framework.
- **ME511: Computational Materials Science** - Developed computational modules, compiled tutorials and documentation, and assisted students with technical and physics-based questions.

Graduate Teaching Assistant/ Research Assistant (West Virginia University)

Aug, 2015 - May, 2022

- Mentored a postdoc and graduate students in performing electronic structure calculations using the **DMFTwDFT**, **PyProcar**, **MechElastic** frameworks and DFT codes (VASP, Abinit, Siesta, Elk, and Quantum Espresso) on HPC clusters.
- **PHYS651, PHYS652: Quantum Mechanics I & II** - Served as the substitute lecturer to cover core topics in quantum mechanics through delivering lectures and leading class discussions.
- **PHYS112: General Physics 2** - Conducted lab sessions and office hours for this calculus-based physics course covering topics in electrostatics, magnetism, and optics receiving recognition as a highly effective TA.
- **ASTR106: Descriptive Astronomy** - Graded homework and exams for this introductory astronomy course.

Instructor (University of Peradeniya)

Dec, 2014 - May, 2015

- **PH211: Vibrations and AC Theory** - Conducted office hours to assist enrolled students. Graded homework and exams.
- **PH230: Quantum Mechanics and Atomic Physics** - Delivered lectures and led class discussions.
- **PH280: General Physics Laboratory I** - Conducted physics lab and tutorial sessions. Graded homework and exams.

Publications

1. Biliriglu, M., Türe, M., Ghita, A., Kotyrov, M., Qin, X., Seyitliyev, D., Phonthiptokun, N., Abdelsamei, M., Chai, J., Su, R., Herath, U., Swan, A. K., Temnov, V. V., Blum, V., So, F., & Gundogdu, K. Unconventional solitonic high-temperature superfluorescence from perovskites. *Nature* **642**, 71–77 (May 2025).
2. Bhat, S. S., Singh, V., Herath, U., Varughese, B., Sankaranarayanan, S. K. R. S., Park, H., & Romero, A. H. Dynamical correlations leading to site and orbital selective Mott insulator transition in hydrogen doped **SmNiO₃**. *Phys. Rev. B* **109**.20, 205124 (May 2024).
3. Lang, L., Tavazde, P., Tellez, A., Bousquet, E., Xu, H., Muñoz, F., Vasquez, N., Herath, U., & Romero, A. H. Expanding PyProcar for new features, maintainability, and reliability. *Computer Physics Communications* **297**, 109063 (2024).
4. Singh, S., Lang, L., Dovale-Farelo, V., Herath, U., Tavazde, P., Coudert, F.-X., & Romero, A. H. MechElastic: A Python library for analysis of mechanical and elastic properties of bulk and 2D materials. *Computer Physics Communications* **267**, 108068 (2021).

- Singh, V., Herath, U., Wah, B., Liao, X., Romero, A. H., & Park, H. DMFTwDFT: An open-source code combining Dynamical Mean Field Theory with various density functional theory packages. *Computer Physics Communications* **261**, 107778 (2021).
- Herath, U., Tavadze, P., He, X., Bousquet, E., Singh, S., Muñoz, F., & Romero, A. H. PyProcar: A Python library for electronic structure pre/post-processing. *Computer Physics Communications* **251**, 107080 (2020).

SUBMITTED

- Abbott, J. W. *et al.* Roadmap on Advancements of the FHI-aims Software Package. *arXiv* (2025).
- Karpov, P., Marek, A., Melson, T., Pöpl, A., Yu, V. W.-z., Hourahine, B., Garcia, A., Dawson, W., Yao, Y., Huhn, W., Moussa, J., Hall, S., Maurer, R., Herath, U., Lion, K., Kokott, S., & Blum, V. Solvers for Large-Scale Electronic Structure Theory: ELPA and ELSI. *arXiv* (2025).
- Herath, U., Bhat, S. S., Singh, V., Lee, A. T., Ngo, A. T., Park, H., & Romero, A. H. Stability and diffusion of oxygen vacancies in LaNiO_3 : a DMFT study. (2025).

IN PREPARATION

- Chakraborty, R., Herath, U., & Blum, V. Hybrid³: A Curated Materials Database for Hybrid Organic-Inorganic Semiconductors.
- Zhang, M.-Y., Herath, U., Merz, F., Yao, Y., Rampp, M., Ren, X., Blum, V., & Scheffler, M. All-electron periodic GW method in numeric atom-centered basis framework: a thorough benchmark.
- Herath, U., Bhat, S., Singh, V., Park, H., & Romero, A. H. Investigating strongly correlated alloys via Dynamical Mean Field Theory.

Selected presentations

- Hybrid³: A Collaborative Platform for Curating Hybrid Perovskite Data. *Duke, NCSU, UNC DMREF Collaboration Meeting, Raleigh, NC. November 6, 2024 (ORAL-INVITED)*
- Database for Hybrid Perovskites: Hybrid³. *NREL CHOICE Meeting, Golden, CO. November 1, 2024 (ORAL-INVITED)*
- A tutorial on PyProcar: A Python library for electronic structure pre/post-processing. *Rutgers University, Piscataway, NJ. June 25, 2021 (ORAL-INVITED)*
- Recent Developments in PyProcar: A Python library for electronic structure pre/post-processing. *APS March Meeting. March 15-19, 2021 (ORAL)*
- DMFTwDFT: An open-source code combining Dynamical Mean Field Theory with various Density Functional Theory packages. *APS Mid Atlantic Section Meeting. December 4-6, 2020 (ORAL)*
- PyProcar: A Python library for electronic structure pre/post-processing. *Carolina Science Symposium, University of California, Merced. November 12-13, 2020 (ORAL)*
- An ab-initio DFT+DMFT study of the effect of oxygen vacancies on structural, electronic and magnetic properties of rare-earth nickelate perovskites (RNiO_3). *APS March Meeting, Boston, MA. March 4-8, 2019 (ORAL)*
- Development of computational methods for the characterization of novel strongly correlated materials. *International Summer School on Computational Quantum Materials, Sherbrooke, Québec, Canada. May 27-June 8, 2018 (POSTER)*
- The Effect of Magnetic Field Line Curvature Scattering on the Rapid Loss of Ring Current Ions. *Geospace Environment Modeling (GEM) conference, Portsmouth, VA. July 2017 (POSTER)*

Fellowships, Grants & Awards

FELLOWSHIPS

ORISE Fellowship

May, 2024

STEM fellowship offered by Oak Ridge National Lab (ORNL) and US Department of Energy (DOE) to conduct materials research in collaboration with the National Energy Technology Laboratory (NETL).

GRANTS

NSF TACC Frontera Pathways - *Research and development of methods for many-body electronic structure theory and applications in FHI-aims*

Dec, 2023 - Dec, 2025

Role: PI

Grant ID: DMR23007

Award: \$64,060.65 (140,557 node-hours on TACC Frontera)

NSF ACCESS Explore - *Benchmarking for "Effect of Molecular Motions on Structural Asymmetry and Electron Spin in 2D Hybrid Halide Perovskites"*

Jan, 2024 - Jan, 2025

Role: Co-PI

Grant ID: CHE240003

Award: \$4456.43 (200,000 ACCESS credits)

NSF XSEDE - *Development of computational methods for electronic structural characterization of strongly correlated materials: from different ab-initio perspectives*

Oct, 2018 - Mar, 2023

Role: Lead contributor (Coordinated a team of graduate students at WVU to secure annual allocations of grant)

Grant ID: DMR140031

Award: \$352,243 (21,636,694 core-hours on PSC Bridges, Bridges2 and 374,942 node-hours on TACC Stampede2)

DOE EPSCOR - *Applications of Nickelate perovskites for neuromorphic computing from electronic structure and Machine Learning*

Sep, 2020 - Aug, 2024

Role: Lead contributor

Grant ID: DOE DE-SC0021375

Award: \$690,000

AWARDS

Sigma Xi Scientific Research Honor Society Associate Membership

Feb, 2025

- Inducted in recognition of scholarly achievements and contributions to the advancement of materials research.

Ovshinsky Travel Award

Feb, 2021

- Awarded by the American Physical Society to fund travel and registration for the APS March Meeting.

Robert T. Bruhn Physics Research Award

May, 2019

- Awarded by the Department of Physics and Astronomy, West Virginia University to extend support to the research effort of a graduate or undergraduate student in the department in nanotechnology and material science.

Office of the Provost Graduate Student Travel Award

2018, 2019

- Awarded by the Office of the Provost, West Virginia University to fund travel to research conferences.

Eberly College of Arts and Sciences Graduate Student Travel Award

2018, 2019

- Awarded by the Eberly College of Arts and Sciences, West Virginia University to fund travel to research conferences.

Referee contributions

- Performed peer review for the *Computer Physics Communications* journal (2022).
- Judged the Virtual Summer Undergraduate Research Symposium organized by the Office of Undergraduate Research in the Mathematics and Physics Category at WVU (July, 2020).

Training & certifications

Emerging Leaders Institute (Duke University)

Jan, 2025 - Mar, 2025

- Developed communication, self-awareness, professional adaptability, project management, interdisciplinary teamwork, and leadership skills through an eight-week professional development program.

Introduction to Data Science in Python (University of Michigan)

Jul, 2025

- Gained experience in data science by manipulating, cleaning, and analyzing data, applying aggregation, merging, pivoting, and statistical analyses to extract insights from complex datasets, while developing familiarity with structured data types (dataframes, graphs, networks, natural language).

Introduction to Machine Learning (Duke University)

Jul, 2025

- Developed a strong foundation in machine learning by implementing models including logistic regression, convolutional neural networks (CNN), and natural language processing techniques (NLP) using PyTorch to address complex problems across diverse domains.

OITE Management Short Course (National Institute of Environmental Health Sciences (NIEHS))

Jun, 2024

- Received training on management principles through small group discussions and case studies.

Responsible Conduct of Research - Staff (CITI/ Duke University)

Jul, 2022

- Received training on core norms, principles, regulations, and rules governing the practice of research including authorship, collaborative research, conflicts of interest, human subjects, and research misconduct.

Leadership & outreach

Co-host - The "Duke Dialogues" Podcast

Mar, 2025 - Present

- Co-hosted a podcast series launched through Duke's ELI program sharing practical advice and resources for navigating graduate and postdoc life.

Council Member - Duke University Postdoc Association (DUPA)

Jul, 2022 - Apr, 2025

- Organized and led events bringing postdocs from different backgrounds together fostering interdisciplinary research collaborations.

Organizer - FHI-aims Developers' and Users' Meeting

Aug 2, 2023 - Aug 4, 2023

- Organized the FHI-aims Developers' and Users' Meeting in Hamburg, Germany, cultivating a vibrant community and promoting knowledge exchange.

Outreach Volunteer - Physics and Astronomy Graduate Student Organization (PAGSO), WVU

Aug, 2015 - May, 2022

- Volunteered in physics outreach programs enhancing public literacy in science across West Virginia, demonstrating skills in public communication, educational outreach, and community service.

Coordinator - Condensed Matter Seminar, Department of Physics and Astronomy, WVU

May, 2018 - May, 2022

- Coordinated weekly seminar sessions bridging sub-disciplines within the condensed matter research community.

Member - Sri Lankan Student's Organization (SLSA), WVU

Jan, 2018 - May, 2022

- Planned events to share Sri Lankan cultural and heritage experiences with the local community and organized a vigil in memory of the Easter Sunday terrorist attack victims in Sri Lanka, building inter-faith cohesion.

President - Astronomy Society of University of Peradeniya

Jan, 2014 - Dec, 2014

- Organized "Water Rocket Challenge 2014", the first-ever water rocket competition at the University of Peradeniya; conducted lectures, planetary observation sessions, and workshops to promote science and astronomy in Sri Lanka.

Junior Editor - Physical Society of University of Peradeniya

Jan, 2013 - Dec, 2013

- Compiled academic articles for the scientific journal "Echo" issued by the society.

Faculty Representative - The Ceylon Drama Society of University of Peradeniya

Jan, 2012 - Dec, 2012

- Organized drama competitions engaging students from various backgrounds.

Volunteer - "Let Them Smile"

Aug, 2010 - Dec, 2014

- Led charity programs focused on improving the education of under-privileged students in rural schools across Sri Lanka.

Volunteer Software Developer - Department of Agriculture, Sri Lanka

Jan, 2010 - Dec, 2010

- Developed a computational framework to diagnose plant disease and suggest treatments in collaboration with the Sri Lanka Department of Agriculture leading to improvements in crop production.